Marc Masgoret

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EDUCATION

Western University

Bachelor of Engineering Science (Mechatronics) + Bachelor of Science (Physics) | September 2024 - May 2028

- GPA 3.9/4.0 | 86.2% cumulative average | Dean's Honour List (2024 - 2026)

WORK EXPERIENCE

Undergraduate Summer Research Internship | Department of Physics | Western University | May - August 2026

- Researched novel high-entropy alloy catalysts for total seawater splitting toward affordable, scalable green hydrogen production, as well as high-entropy oxide semiconductors for photocatalytic hydrogen evolution, under the supervision of Dr. Amir Mirzaei.
- Applied AI and machine learning to process large datasets and identify promising metal compositions.
- Fabricated catalysts and substrate nanotubes through electrodeposition and hydrothermal synthesis, characterized samples with techniques such as X-ray diffraction and UV spectroscopy, and evaluated performance and durability through potentiostat testing.
- Presented highly competitive results at Western's 2026 USRI Research Conference, now being carried forward for publication.

Midtown Medical Family Physician Practice | Toronto, ON | Summers 2020 - 2024

- Built and maintained the clinic website (50,000+ visits), hosting medical information and patient forms.
- Coordinated specialist follow-ups on outstanding patient results, supported front desk administration for the physicians, and assisted with office renovation during a practice merger.

ENGINEERING EXPERIENCE

Western Engineering FIRST Club

VP Combat Robotics | August 2026 – Present

- Leading Western's combat robotics program, driving technical design decisions, managing team leads, and overseeing budgets and timelines across two parallel builds to test design approaches before entering competitions against other universities.

Team Member | November 2025 – April 2026

- Independently built an autonomous maze-solving robot using ESP32s, ultrasonic and IR sensors, IMUs, and encoders.

Humanoid AI Teaching Assistant Robot | **Arm Subsystem Lead** | **Manufacturing Lead (Full Robot)** | March - September 2026

- Part of a student team commissioned by Dr. Mark Baker to design and build a life-sized (waist-up) humanoid robot as an AI-powered teaching assistant to enhance lecture delivery and student engagement.
- Led the full mechanical and electrical development of the arm subsystem team, from design through fabrication and integration.
- Assisted in the fabrication and assembly of the neck, face, and torso subsystems.

Western Engineering Baja Racing | **Mechanical Design and Assembly** | November 2024 - Present

- Active member and one of the lead designers in the powertrain subsystem.
- Designed and manufactured a split drive shaft using U-joints to allow for a lowered engine and driver's seat while optimizing shaft material and proportions to minimize weight.
- Designed sprag bearing wheel hubs to eliminate the front transmission and redesigned wheel rims to reduce weight.

Quantum Logistics Capstone | **Contributor** | January - March 2026

- Contributed data collection and analysis to a capstone project comparing quantum and classical route-planning methods for large-scale delivery optimization.

Engineering in Global Context | **Student Exchange** | Universitat Autonoma de Barcelona | May - June 2025

- Gained international engineering experience emphasizing practical industrial large-scale applications in oil and gas (Repsol), recycling, and water filtration plants.

TECHNICAL SKILLS AND CERTIFICATIONS

Technical Skills:

- Design: SolidWorks, Autodesk Inventor, Altium Designer, Microcap, 3D printing slicers
- Tools: C++, C, Origin, MATLAB, Granta EduPack, Simulink, Java, Python, Excel, Git, EIS Analyser, VersaStudio, Microsoft Office
- Manufacturing and Prototyping: Soldering, circuit debugging (multimeter & oscilloscope), mechanical assembly, 3D printing and rapid prototyping, basic machining

Certifications:

- Certified SolidWorks Associate (CSWA), Advanced Machine Shop Training, Introduction to Welding Training, Machine Learning with Python (A-Z Course), Ontario Lifeguard and Swim Instructor Certificates

COMPETITIONS AND ACTIVITIES

Competitions:

- Placed 2nd in Western's mechatronics second-year design competition, building a two-stage autonomous system with a collection robot following a preplanned route covering 30 m^2 and a vibration V-funnel sorter that indexes and sorts erasers by colour. | January - April 2026
- Placed 1st in Western's First Year Engineering Design Competition. Worked for months to design an open-source, 3D printable, affordable 3D printer enclosure with an active filtration system. | January - April 2025

Activities:

- Rock climbing (competed nationally), won basketball intramurals, weightlifting, long distance running, intramural soccer, drums